

Alexander Martin

amart233@jhu.edu • [GitHub](#) • [Scholar](#) • [Website](#)

RESEARCH INTERESTS

My research focuses on multimodal retrieval and generation, specifically towards advancing efficient and scalable collaborative multimodal search and reasoning.

EDUCATION

Johns Hopkins University <i>Ph.D. in Computer Science</i> <i>Advisor: Dr. Benjamin Van Durme</i>	Baltimore, Maryland Expected 2029
University of Rochester <i>B.S. in Computer Science; Highest Honors in Research</i> <i>Advisor: Dr. Aaron Steven White, Dr. Jiebo Luo</i> <i>Thesis: Human-Centric Event Representations in Documents and Videos</i>	Rochester, New York May 2024

SELECT HONORS & AWARDS

National Science Foundation, Graduate Research Fellowship	2024 – 2029
Computing Research Association, Outstanding Undergraduate Research Award (HM)	2024
University of Rochester, Charles L. Newton Prize	2024

INDUSTRY RESEARCH

Amazon, Fire TV <i>Researcher; AI2AI Award; PI: Dr. Mahsa Yarmohammadi, Amazon PI: Dr. Giovanni Seni</i> Researched efficient multimodal retrieval, regularly presenting findings to collaborators at Fire TV.	2025 – 2026
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RESEARCH EXPERIENCE

Human Language Technology Center of Excellence <i>Ph.D. Researcher; Advised by Dr. Benjamin Van Durme</i>	2024 – Present
Formal And Computational Semantics Lab <i>Undergraduate Researcher; Advised by Dr. Aaron Steven White</i>	2022 – 2024
Visual Intelligence & Social Multimedia Analytics Lab <i>Undergraduate Researcher; Advised by Dr. Jiebo Luo</i>	2022 – 2024
Environmental Protection Agency <i>Research Intern; Advised by Dr. Andrea Kirk</i>	Summer 2022 – Fall 2022
Rochester Human Computer Interaction Lab <i>Research Assistant; Advised by Dr. Ehsan Hoque</i>	2022 – 2024

PUBLICATIONS

Selected Publications: Multimodal Retrieval, Efficiency, RAG

- [1] **A. Martin**, et al. “Knowledge Pull Requests for Continual Document Authoring” ([ArXiv](#) 2026)
- [2] **A. Martin**, et al. “Seeing Through the MiRAGE: Evaluating Multimodal Retrieval Augmented Generation” ([EMNLP](#) 2026)
- [3] **A. Martin**, et al. “WikiVideo: Article Generation from Multiple Videos” ([ACL](#) 2026)
- [4] H. Qin*, **A. Martin***, et al. “Multi-Vector Index Compression in Any Modality” ([SIGIR](#) 2026)
- [5] A. Reddy*, **A. Martin***, et al. “Video-ColBERT: Contextualized Late Interaction for Text-to-Video Retrieval” ([CVPR](#) 2025)

All Others: [Google Scholar](#)

TALKS

A Three-Stage Pipeline for Video Retrieval Augmented Generation [talk] The 2nd Workshop on Multimodal Augmented Generation via Multimodal Retrieval	July 2026 Oral
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Knowledge Intensive Multimodal Retrieval Augmented Generation [talk] University of Maryland, Baltimore County; PPR Seminar	November 2025 Seminar
Discovering and Understanding Events in Videos in Real Time [slides] MacroXStudio (macrox.ai)	October 2025 Invited Talk
On Demand Article Generation of Events in Real Time [slides] University of Rochester, Department of Computer Science	April 2025 Seminar
MARE: Automatic Modality-Agnostic Report Evaluation [slides] Eval4Rag Workshop; European Conference on Information Retrieval	April 2025 Oral
Understanding Events in Multimodal Data Through Question Answering [slides] 4 th Workshop on Processing and Evaluating Event Representations	April 2025 Oral
FAMuS: Frames Across Multiple Sources [slides] 3 rd Workshop on Processing and Evaluating Event Representations	April 2024 Oral
Jurassic World Remake [slides] Main Conference; ACM Multimedia	October 2023 Oral

SERVICE

MAGMaR: Workshop on Multimodal Augmented Generation via Multimodal Retrieval

Organizing Committee; *ACL 2026*
Program Committee; *ACL 2025*

Students Mentored

Charles Weng (**Initial Placement:** TikTok Research Scientist)
Dengjia Zhang (**Initial Placement:** University of Pittsburgh, Ph.D.)
Hanxiang Qin (**Initial Placement:** MLE at Kuaishou)
Katherine Guerrero (**current:** Johns Hopkins University, B.S. 2027, *Pistritto Fellowship*)

Community Service

North American Computational Linguistics Open Competition (NACLO) Competition Organizer	2024 – Present
STEM Initiative Education Mentor	2021 – 2024
University of Rochester: Residential Life Resident Advisor	2021 – 2024

Teaching

Introduction to Artificial Intelligence (University of Rochester)	Spring 2023
Data Structures and Algorithms (University of Rochester)	Spring 2022, Fall 2022
Introduction to Computer Science (University of Rochester)	Fall 2021

OTHER HONORS & AWARDS

University of Rochester, Senior Research Award	2024
University of Rochester, Research Presentation Grant (x2)	2023, 2024
University of Rochester, Dean's Award in Engineering and Mathematics (x2)	2023, 2024
University of Rochester, River Campus Libraries Dataset Grant	2023
University of Rochester, Residential Life Best Program of the Year	2023
University of Rochester, Make It Happen Grant	2022